

Brain Tumor Detection & Classification

from MRI Scans using Image Processing & Deep Learning

A complete image processing pipeline for automated brain tumor classification

M.H.Nasrullah

Higher National Diploma in Computer Science with Artificial Intelligence
NIBM — The City University | KIC-HNDCSAI-252F-031

Dataset: Brain Tumor MRI Dataset — 7,200 images across 4 classes

Best Model Accuracy: 90.56% (EfficientNetB0 Transfer Learning)

Tools: Python | OpenCV | TensorFlow / Keras | Scikit-learn | XGBoost

Table of Contents

1. Project Overview & Objective
2. Dataset Description
3. Image Processing Pipeline
 - 3.1 Preprocessing — Gaussian Blur
 - 3.2 Enhancement — Histogram Equalization
 - 3.3 Edge Detection — Canny Algorithm
 - 3.4 Morphological Operations
 - 3.5 Tumor Segmentation
 - 3.6 Contour Extraction & Visualization
4. Feature Extraction
5. Classification Models
 - 5.1 Random Forest
 - 5.2 XGBoost
 - 5.3 CNN — Original Architecture
 - 5.4 CNN — EfficientNetB0 Transfer Learning (Improved)
6. Results & Evaluation
7. Prediction Pipeline Visualization
8. Discussion
9. Conclusion

1. Project Overview & Objective

Brain tumors are one of the most life-threatening medical conditions. Early and accurate detection directly influences treatment decisions and patient survival rates. MRI (Magnetic Resonance Imaging) is the primary diagnostic tool used by neurologists to examine the brain, but manually reviewing and classifying hundreds of MRI scans is time-consuming, subjective, and prone to human error — particularly in high-volume clinical settings.

This project builds a complete automated image processing and classification pipeline for brain MRI scans. The system classifies each scan into one of four categories:

- **Glioma** — A fast-growing tumor originating in glial cells, often irregular in shape and difficult to distinguish from other tumors.
- **Meningioma** — A typically slow-growing tumor arising from the meninges (protective membranes around the brain).
- **Pituitary Tumor** — A growth in the pituitary gland at the base of the brain, often identifiable by its fixed anatomical location.
- **No Tumor** — A healthy brain scan with no abnormal growth present.

The emphasis of this project is on the image processing side — understanding and applying techniques such as noise removal, contrast enhancement, edge detection, morphological operations, and segmentation — before applying machine learning and deep learning classifiers to achieve the final prediction.

The final system achieved 90.56% classification accuracy using EfficientNetB0 transfer learning, improved from an initial 77.00% with a basic CNN.

2. Dataset Description

The dataset used is a widely referenced brain tumor MRI classification dataset containing a total of 7,200 images split across training and testing sets:

Training Set: 5,600 images (1,400 per class)
Testing Set: 1,600 images (400 per class)
Classes: Glioma | Meningioma | Pituitary | No Tumor
Image Format: Grayscale MRI scans (JPG/PNG), variable sizes
MRI Views: Axial, Coronal, and Sagittal perspectives

The dataset is balanced — each class has an equal number of images in both the training and testing sets, which means no class imbalance correction was necessary. The images come from different MRI scanners and include various scan orientations (axial = top-down view, coronal = front view, sagittal = side view), which adds natural variety and makes the classification task more realistic.

Sample MRI Images — 3 per Class

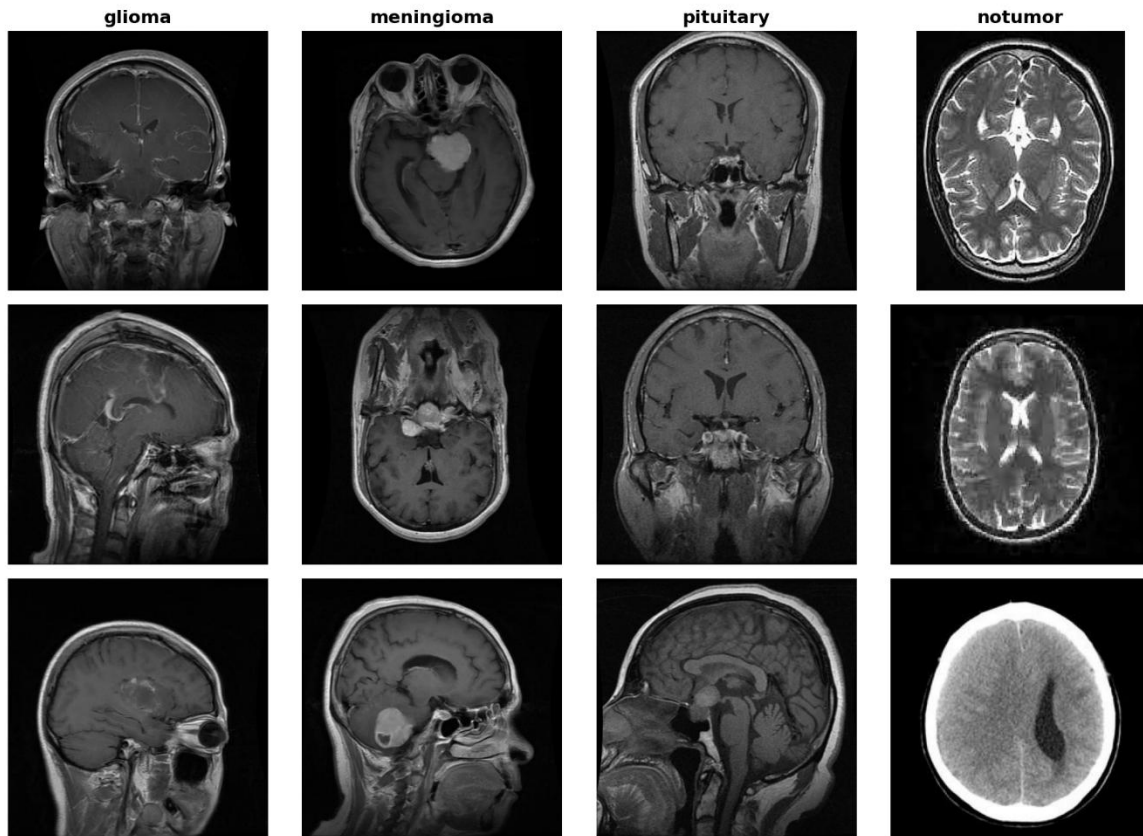


Figure 1: Sample MRI images from the dataset — 3 images per class showing the variety of scan orientations

3. Image Processing Pipeline

Before any classification is attempted, each MRI image passes through a series of image processing steps. These steps serve two purposes: (1) improving the visual quality and clarity of the images, and (2) extracting meaningful information about the tumor region for use in feature-based classifiers.

The complete pipeline is:

- **Gaussian Blur** → Histogram Equalization → Edge Detection → Morphological Operations → Segmentation → Contour Extraction → Feature Extraction → Classification

3.1 Preprocessing — Gaussian Blur

Gaussian Blur is applied as the first preprocessing step to reduce noise in the MRI scans. MRI images inherently contain small intensity fluctuations (often called "MRI noise" or Rician noise) that are not clinically meaningful but can interfere with downstream edge detection and segmentation algorithms.

A 5×5 Gaussian kernel was used. The kernel size was chosen because it provides enough smoothing to reduce noise while still preserving the important structural boundaries between brain tissue and tumor regions. A larger kernel would over-smooth and lose edge information; a smaller kernel would leave too much noise.

Additionally, a second, smaller 3×3 Gaussian blur is applied specifically before the Canny edge detection step, to reduce noise-induced false edges at that stage.

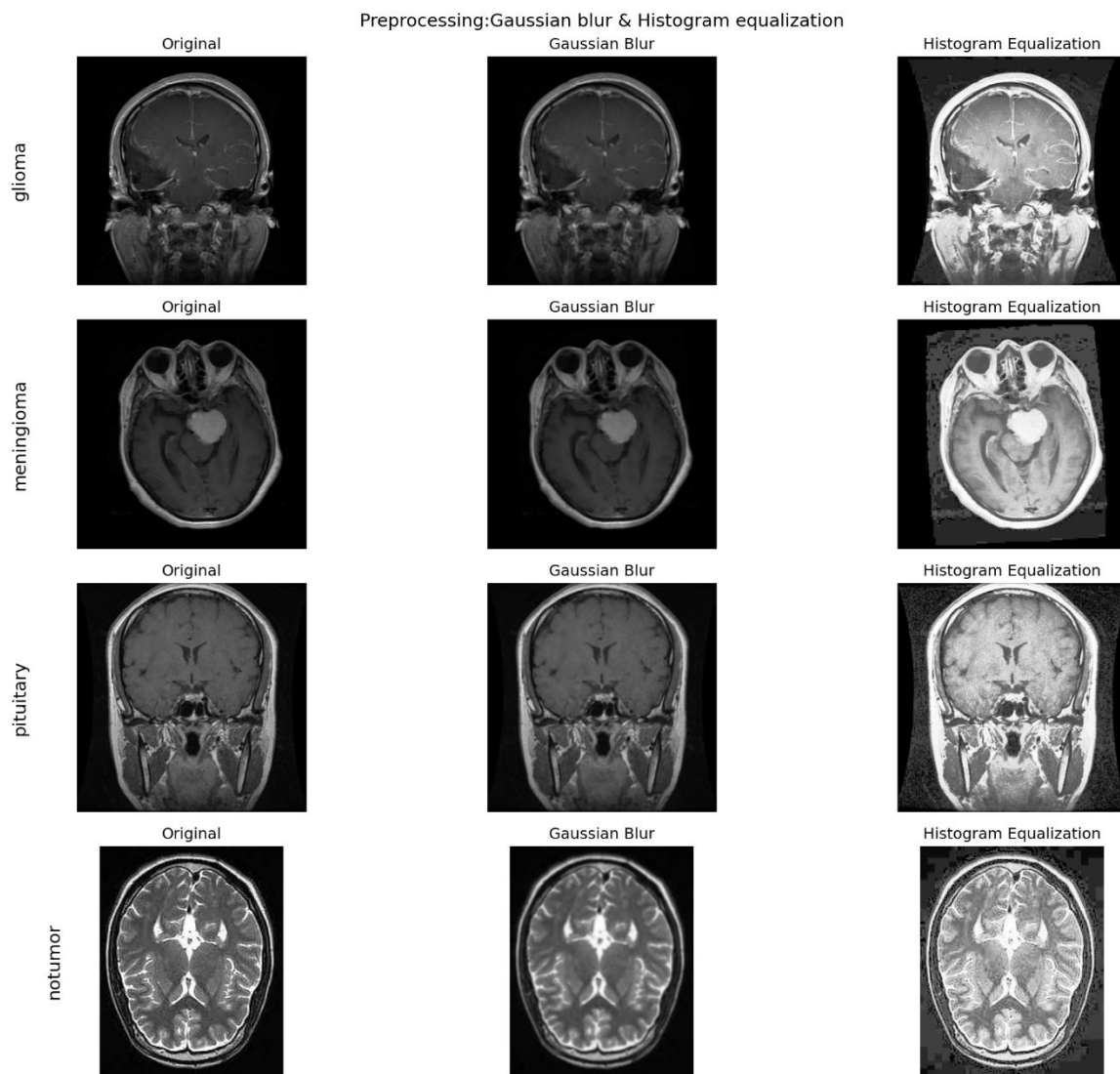


Figure 2: Original | Gaussian Blur | Histogram Equalization — applied to all four tumor classes

The visual effect of Gaussian blur is subtle but important — the images look slightly smoother and the fine grain texture noise is reduced, making the larger structural boundaries more prominent.

Individual class results after Gaussian blur:

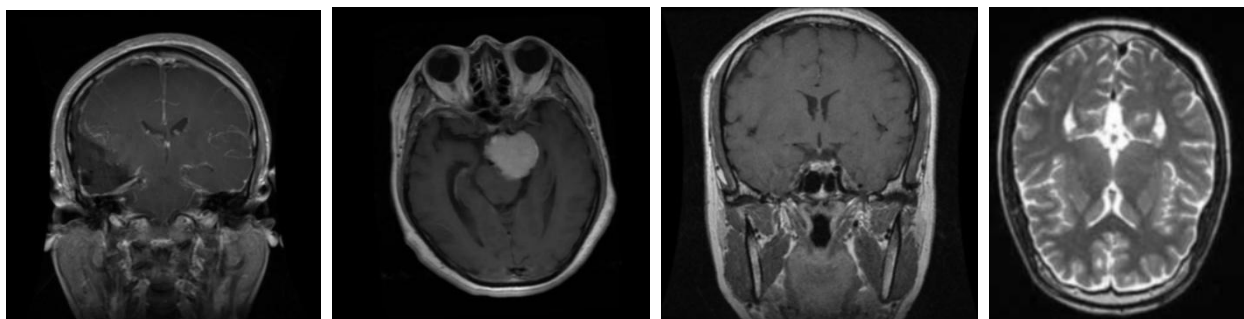


Figure 3: Gaussian blur applied to individual class samples — Glioma | Meningioma | Pituitary | No Tumor

3.2 Enhancement — Histogram Equalization

Histogram Equalization is applied to improve the contrast of the MRI images. In many MRI scans, the pixel intensity values are clustered in a narrow range — typically the darker regions dominate because the background of an MRI is black, and only the brain tissue contributes meaningful intensities. This makes it harder to visually distinguish tumor tissue from surrounding brain matter.

The equalization is performed on the Y channel (luminance) after converting the image from RGB to YCrCb color space. This approach is more controlled than equalizing all three channels independently, as it only adjusts brightness while preserving the color information (important if the original images have any color encoding).

The result is a more spread-out intensity distribution, as confirmed by the histogram plots below, where the original histograms show heavy concentration near zero (dark pixels), while the post-equalization histograms show a much more uniform spread across 0–255.

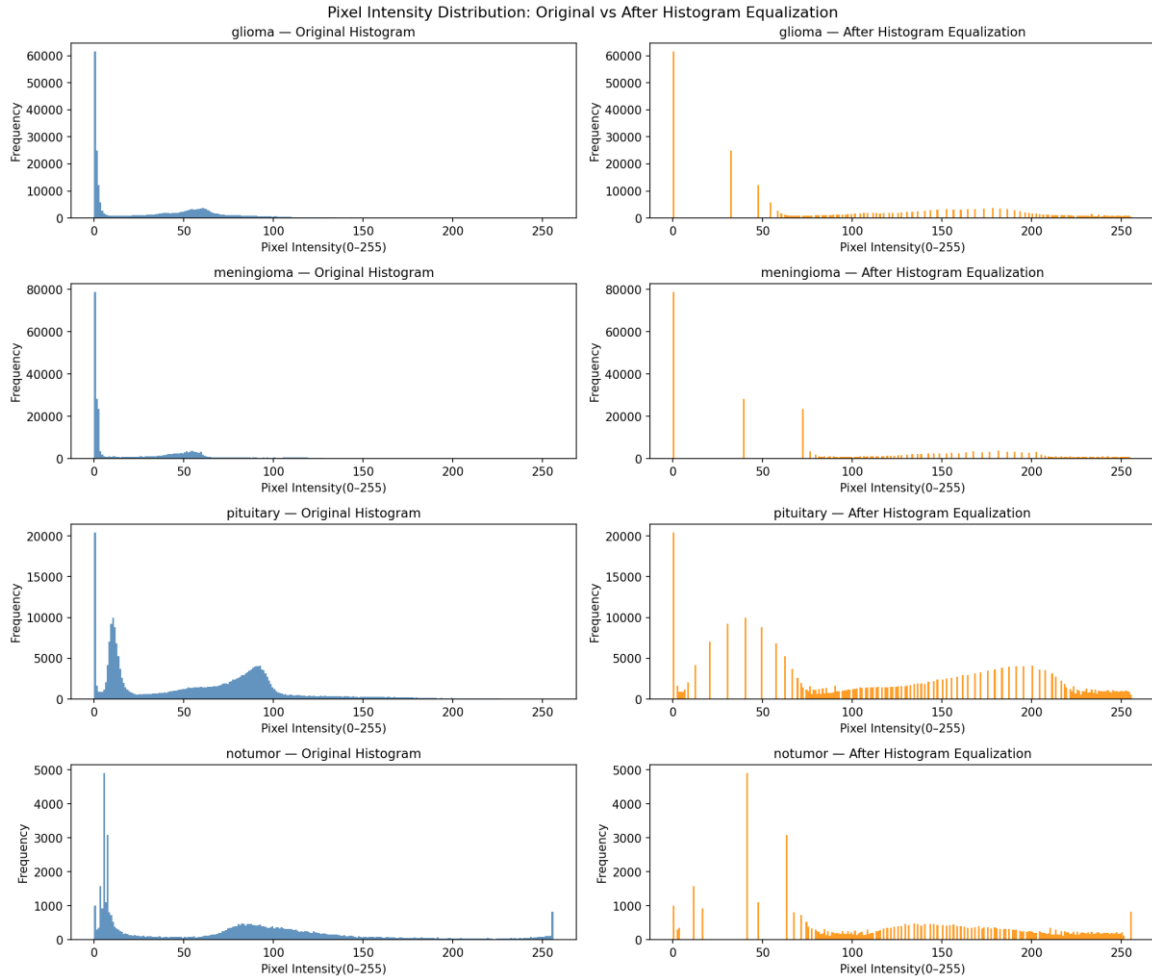


Figure 4: Pixel intensity distributions before and after Histogram Equalization for all four classes

Individual class results after Histogram Equalization:

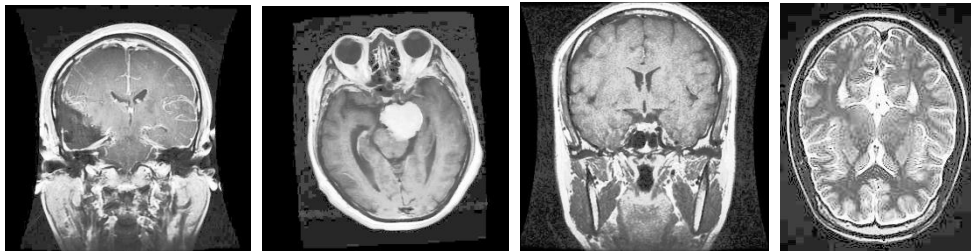


Figure 5: Histogram Equalization applied to individual class samples — enhanced contrast visible across all classes

3.3 Edge Detection — Canny Algorithm

Canny edge detection is applied to identify the structural boundaries within each MRI scan. In the context of brain tumor analysis, edges correspond to transitions between different tissue types — the boundary between the tumor mass and surrounding brain tissue, the skull boundary, ventricles, and other anatomical structures.

The Canny algorithm was chosen because it is a multi-stage algorithm that produces clean, thin edge lines with minimal noise. It works by: (1) smoothing the image with a Gaussian filter, (2) computing the image gradient to find intensity changes, (3) applying non-maximum suppression to thin the edges, and (4) using hysteresis thresholding to select only strong, connected edges.

Algorithm Parameters:

Pre-blur kernel: 3×3 Gaussian (applied before Canny to reduce noise-induced false edges)
 Low threshold: 50 (edges weaker than this are discarded)
 High threshold: 150 (edges stronger than this are always kept)
 Hysteresis: edges between 50–150 are kept only if connected to a strong edge

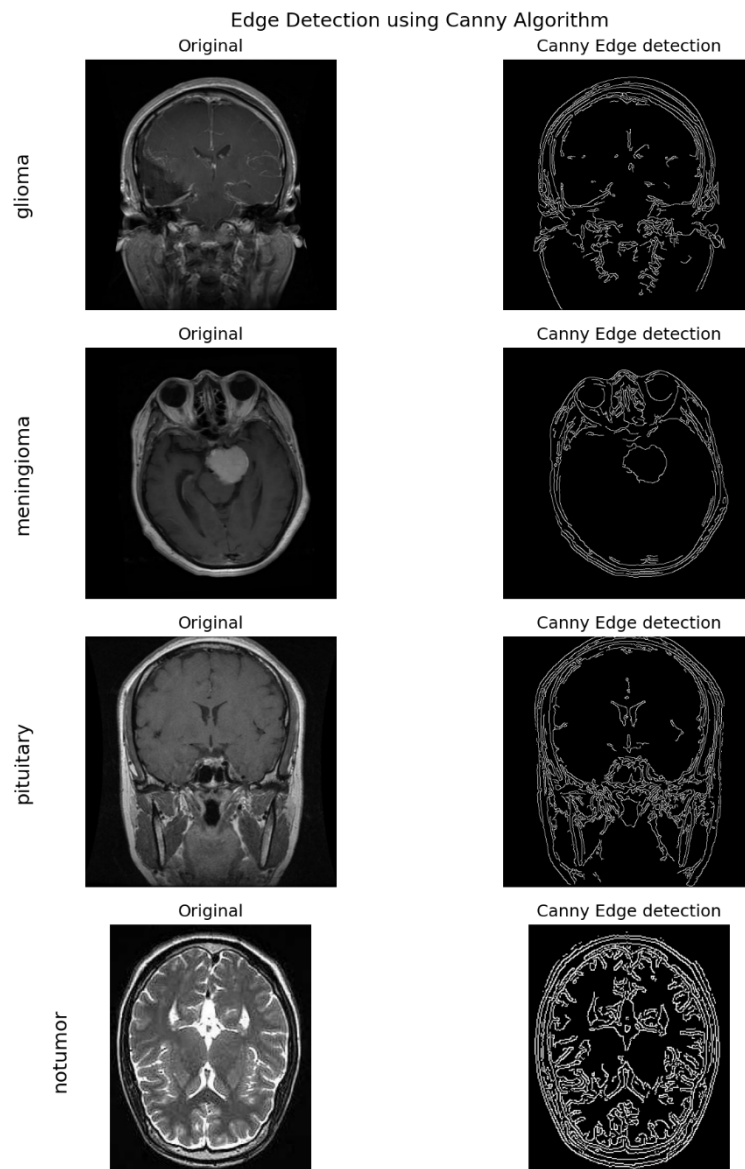


Figure 6: Canny edge detection results — Original (left) and edge map (right) for all four classes

Each class shows distinctly different edge patterns. Meningioma scans show a clear circular edge around the tumor mass. Glioma scans show irregular, scattered edges reflecting the diffuse nature of the tumor. No Tumor scans show dense, uniform edges reflecting normal brain gyri and sulci structure.

Individual class edge detection results:

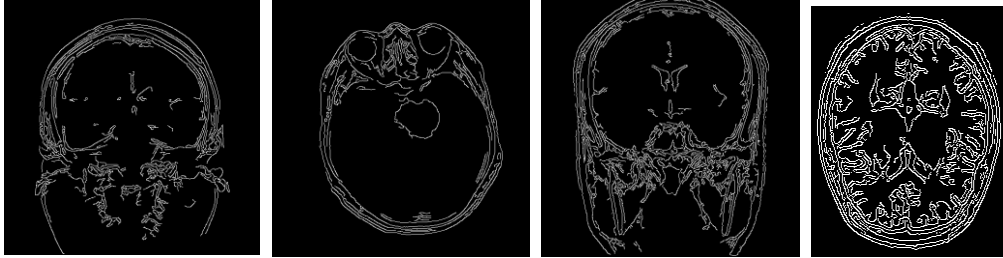


Figure 7: Canny edge maps per class — Glioma | Meningioma | Pituitary | No Tumor

3.4 Morphological Operations

Morphological operations are fundamental image processing techniques that work on binary or grayscale images based on a structuring element (a predefined shape used as a probe). In this project, four morphological operations are applied to the grayscale MRI images using a 5×5 elliptical structuring element.

These operations are important because MRI binary masks (produced by thresholding) often contain noise and gaps that need to be cleaned before features can be reliably extracted.

Erosion

Erosion shrinks all bright (white) regions in the image by removing pixels at the boundaries of bright regions. Any bright region smaller than the structuring element is completely removed. This is useful for eliminating small noise pixels and thin protrusions.

Dilation

Dilation expands all bright regions by adding pixels around the boundaries. Small holes within bright regions get filled in. This is used to close small gaps and connect nearby regions.

Opening (Erosion followed by Dilation)

Opening removes small noise objects from the background while preserving the shape and size of larger objects. It is particularly useful for cleaning up the binary masks produced after Otsu thresholding in the segmentation step.

Closing (Dilation followed by Erosion)

Closing fills small holes and gaps within the foreground objects while preserving their overall shape. This helps complete tumor regions that have small internal dark spots.

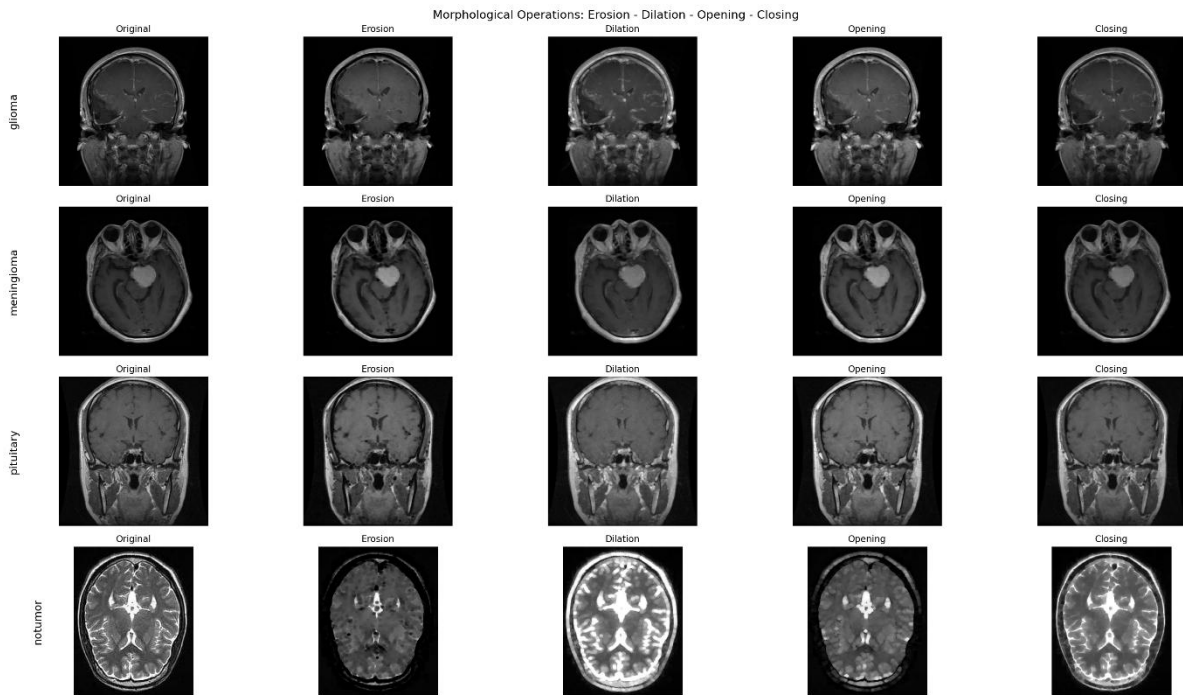


Figure 8: Morphological operations applied to all four classes — Original | Erosion | Dilation | Opening | Closing

3.5 Tumor Segmentation

Tumor segmentation is the process of isolating the tumor region specifically from the rest of the brain scan. This was one of the most challenging steps in the project because the goal is to detect the tumor mass, not the skull, not the brain outline, and not any other brain structure.

The segmentation pipeline consists of three stages:

- **Stage 1 — Otsu Thresholding:** Otsu's method automatically determines the optimal threshold value that best separates the bright foreground (tumor tissue) from the darker background. The threshold is computed by maximizing the variance between the two resulting pixel classes. This is better than using a fixed threshold because MRI scan brightness varies between images.
- **Stage 2 — Morphological Opening:** After thresholding, the binary mask is cleaned using morphological opening (2 iterations, 5×5 ellipse kernel) to remove small noise blobs that are not part of the tumor.
- **Stage 3 — Contour Area Filtering:** All contours in the cleaned mask are found and sorted by area. The largest contour is almost always the skull or overall brain outline and is deliberately skipped. Contours with area between 1% and 30% of the largest contour area are retained as tumor candidates. This filtering step was critical to avoid segmenting the whole brain instead of the tumor.

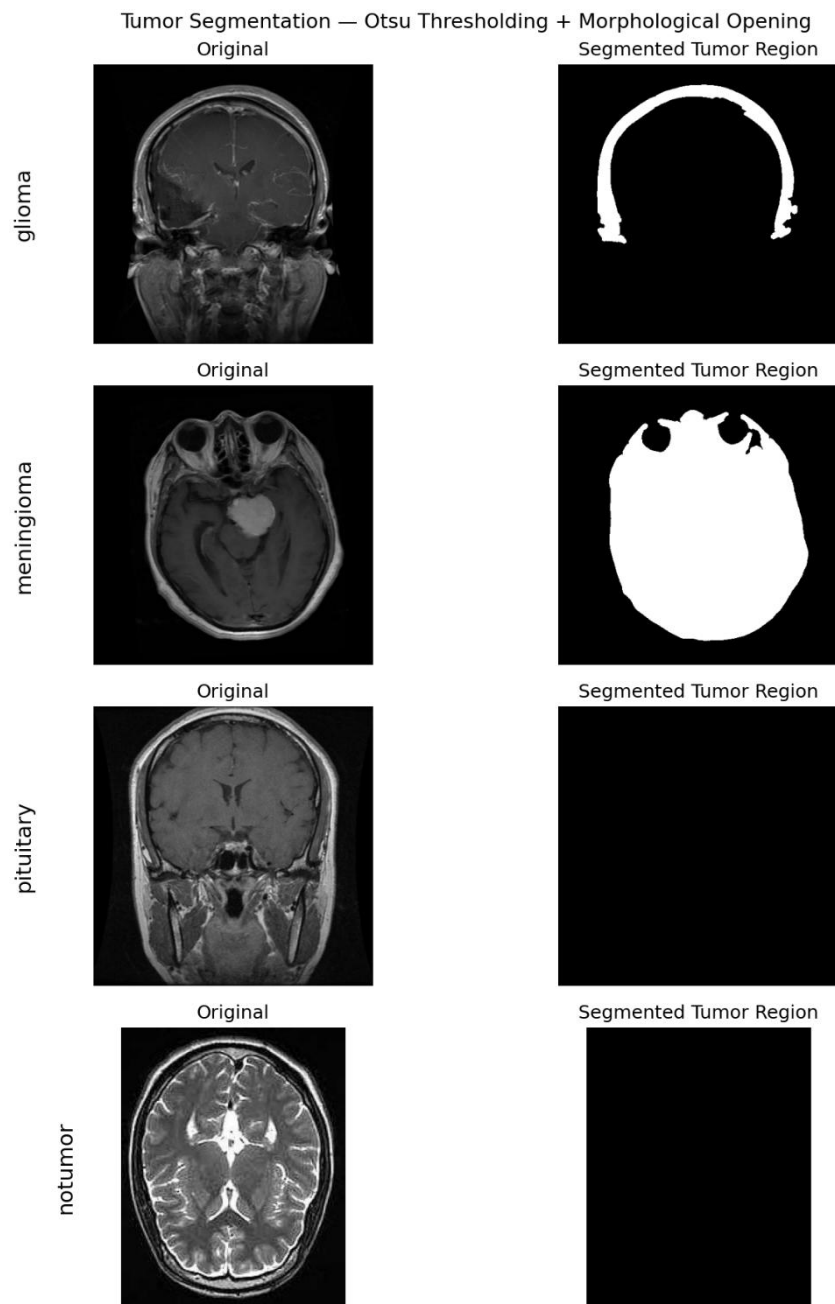


Figure 9: Tumor segmentation results — Original (left) and segmentation mask (right) for all four classes

The segmentation results vary by class. Meningioma scans tend to produce cleaner masks because meningiomas are typically well-defined, bright, circular tumors. Glioma and Pituitary scans are harder because the tumor intensity can be closer to surrounding tissue. No Tumor scans correctly produce empty or near-empty masks since there is no bright abnormal region to detect.



Figure 10: Segmentation masks — Glioma (left) and Meningioma (right). The Meningioma mask is cleaner due to the well-defined bright tumor mass.

3.6 Contour Extraction & Visualization

After segmentation produces a binary tumor mask, contours are extracted from the mask and drawn on top of the original MRI image. This step serves as both a visual verification tool and produces the shape features used by the traditional classifiers.

The visualization uses two overlays:

- **Green line:** The detected tumor contour boundary drawn directly on the original MRI image.
- **Blue rectangle:** A bounding box drawn around the largest detected contour, showing the spatial extent of the tumor candidate.

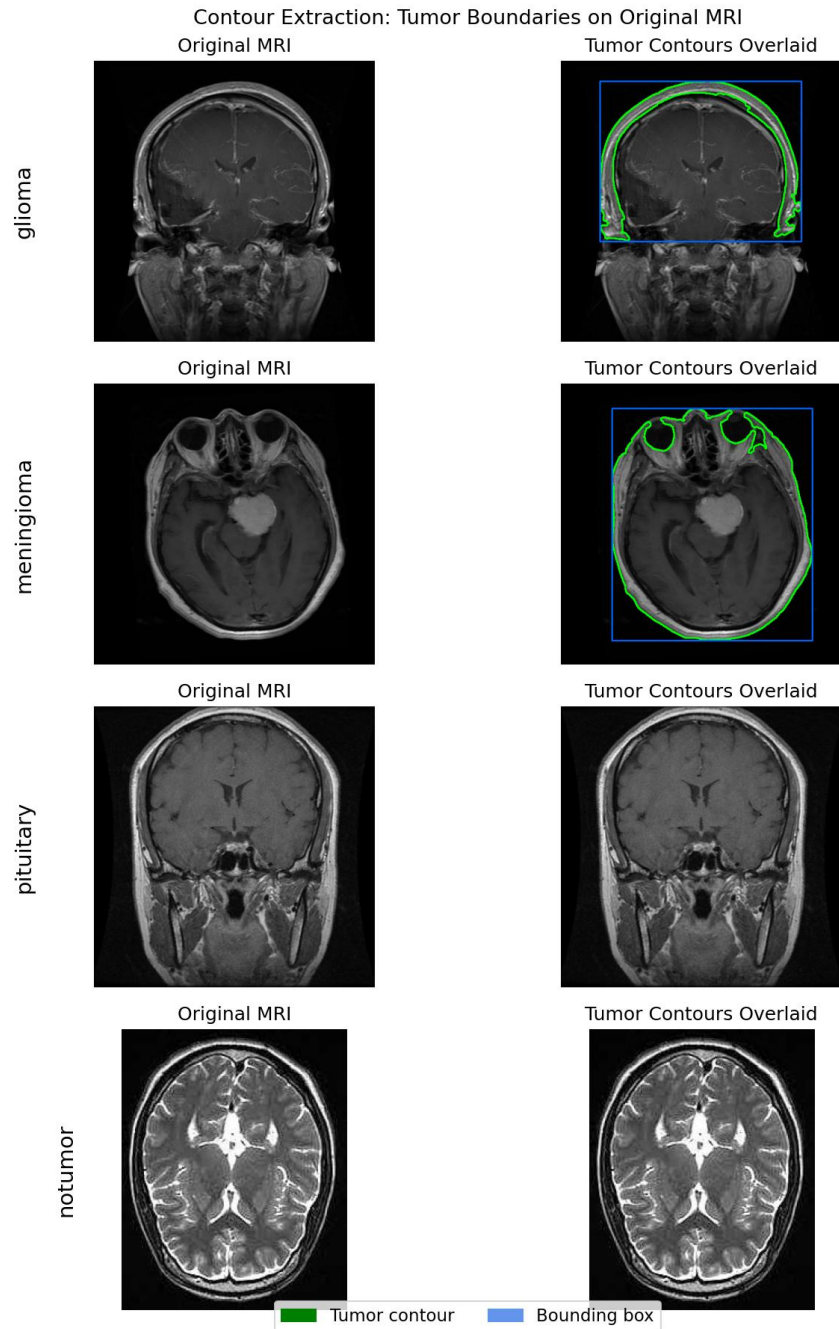


Figure 11: Contour extraction and visualization for all four classes — green contour line and blue bounding box overlaid on original MRI

The contour overlay is most informative for Meningioma, where the green contour clearly traces the circular tumor boundary. For Glioma, the contours tend to follow larger structural boundaries. For No Tumor and Pituitary scans with no detected tumor region, no contours or minimal contours are shown.

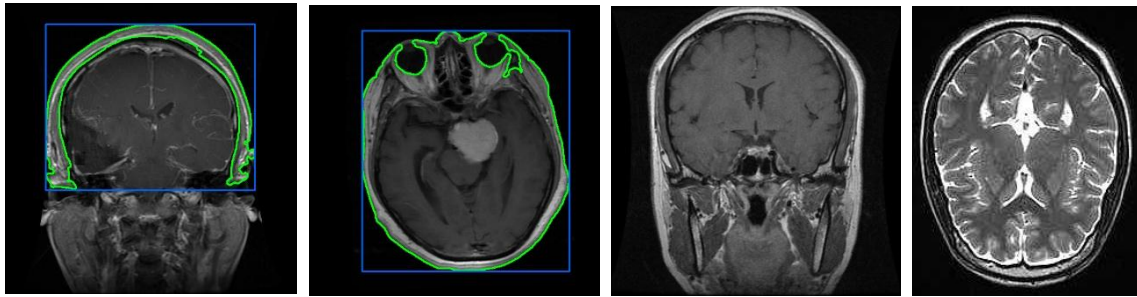


Figure 12: Individual contour visualization per class — Glioma | Meningioma | Pituitary | No Tumor

4. Feature Extraction

For the traditional machine learning classifiers (Random Forest and XGBoost), eight numerical features are extracted from each image. These features describe the shape and intensity properties of the segmented tumor region and are used as input to the classifiers instead of raw pixel values.

Shape Features (from the segmentation mask contour):

- **Area:** The total number of pixels within the largest detected contour. Larger tumors produce higher area values.
- **Perimeter:** The total length of the contour boundary. Irregular tumors (like glioma) tend to have larger perimeters relative to their area.
- **Aspect Ratio:** The ratio of the bounding box width to height (W/H). A value near 1.0 indicates a roughly circular/square tumor; values further from 1.0 indicate elongated shapes.
- **Circularity:** Computed as $4\pi \times \text{Area} / \text{Perimeter}^2$. A perfect circle has circularity = 1.0. Meningiomas, being more circular, tend to have higher circularity values than gliomas.

Intensity Features (from pixels inside the mask):

- **Mean Intensity:** The average pixel brightness within the tumor region. Bright tumors score higher.
- **Standard Deviation:** The spread of pixel intensities within the tumor. Higher values indicate more heterogeneous (varied) tissue texture.
- **Minimum Intensity:** The darkest pixel within the segmented region.
- **Maximum Intensity:** The brightest pixel within the segmented region.

These 8 features are extracted from all 5,600 training images and saved to a CSV file. The dataset is then split 80%/20% (4,480 training, 1,120 testing) for use in Random Forest and XGBoost training. While these features capture some useful characteristics, they are inherently limited by the quality of the segmentation — poor segmentation produces unreliable feature values, which is one reason why the feature-based classifiers underperform compared to the CNN.

5. Classification Models

Four classification approaches were implemented and compared. The progression from traditional machine learning to deep learning clearly illustrates the advantage of learning directly from image pixels over manually engineered features.

5.1 Random Forest

Random Forest is an ensemble learning method that builds 200 independent decision trees, each trained on a random subset of the training data and features. The final prediction is made by majority vote across all trees.

Algorithm: Random Forest Classifier

n_estimators: 200 trees | random_state: 42 | Input: 8 features per image
Training: 4,480 images (80% of 5,600) | Test: 1,120 images (20%)
Accuracy: 65.45%

The Random Forest performed reasonably well given only 8 input features, but its accuracy is limited by the quality of the segmentation-derived features. Images where the segmentation fails (empty or incorrect mask) contribute zero or incorrect features, which directly hurts performance.

5.2 XGBoost

XGBoost (Extreme Gradient Boosting) is a boosting method that builds trees sequentially, with each new tree correcting the errors of the previous ones. It often outperforms Random Forest on tabular data.

Algorithm: XGBoost Classifier

n_estimators: 200 | eval_metric: mlogloss | Input: 8 features per image
Training: 4,480 images | Test: 1,120 images
Accuracy: 64.82%

XGBoost performed slightly below Random Forest (64.82% vs 65.45%), which is unusual since XGBoost typically performs better on tabular data. This suggests the features themselves are the bottleneck — both models are limited by the same 8 imperfect segmentation-based features.

5.3 CNN — Original Architecture

A Convolutional Neural Network was built from scratch to learn directly from raw pixel values, bypassing the feature extraction and segmentation steps entirely. The architecture consists of three convolutional blocks followed by a fully connected classifier.

Architecture: 3× (Conv2D → MaxPooling2D) → Flatten → Dense(128) → Dropout(0.5) → Dense(4)
Input size: 128×128×3 | Batch size: 32 | Epochs: 10
Optimizer: Adam | Loss: Categorical Crossentropy
Augmentation: rotation, shift, zoom, horizontal flip
Accuracy: 77.00%

The original CNN achieved 77% accuracy, significantly outperforming both feature-based models. This confirms that raw pixel patterns contain far more discriminative information than 8 hand-crafted features. However, the relatively shallow architecture and only 10 training epochs limited the potential accuracy.

5.4 CNN — EfficientNetB0 Transfer Learning (Improved)

To significantly improve accuracy, the original CNN was replaced with an EfficientNetB0 backbone using transfer learning. EfficientNetB0 is a state-of-the-art convolutional neural network pre-trained on ImageNet (1.2 million images, 1,000 classes). It already knows how to detect low-level features like edges, textures, and shapes — exactly the types of features needed for MRI analysis.

A critical issue was discovered during initial testing: the original data generator used `rescale=1./255` for pixel normalization, but `EfficientNetB0` requires its own specific preprocessing function (`preprocess_input` from `tensorflow.keras.applications.efficientnet`). Using the wrong normalization caused the model to receive completely wrong pixel values, resulting in only 37% accuracy — worse than random guessing. After fixing this, the model trained correctly.

```
Base model: EfficientNetB0 (ImageNet pre-trained weights)
Input size: 224×224×3 (increased from 128×128 for better feature resolution)
Preprocessing: preprocess_input (not rescale=1./255)
Top layers: GlobalAveragePooling2D → BatchNorm → Dense(256) → Dropout(0.4) → Dense(128) → Dropout(0.3) → Dense(4, softmax)
Phase 1: Freeze base, train top layers (Adam lr=1e-3, 15 epochs)
Phase 2: Unfreeze top 50 EfficientNet layers, fine-tune (Adam lr=1e-4, 20 epochs)
Callbacks: EarlyStopping (patience=6) | ReduceLROnPlateau (factor=0.3, patience=3)
Final Accuracy: 90.56%
```

The two-phase training strategy is important. In Phase 1, the `EfficientNet` base layers are frozen and only the new top classifier layers are trained. This allows the top layers to reach a reasonable performance before fine-tuning begins. In Phase 2, the last 50 layers of `EfficientNet` are unfrozen and the whole network is trained together with a very low learning rate (`1e-4`) to avoid destroying the pre-trained weights.

6. Results & Evaluation

6.1 Overall Model Comparison

Model	Training Data	Test Images	Accuracy	Improvement
Random Forest	4,480 (features)	1,120	65.45%	Baseline
XGBoost	4,480 (features)	1,120	64.82%	-0.63%
CNN (Original)	5,600 (raw pixels)	1,600	77.00%	+11.55%
CNN + EfficientNetB0	5,600 (224×224)	1,600	90.56%	+25.11%

Table 1: Complete model accuracy comparison — all four classifiers

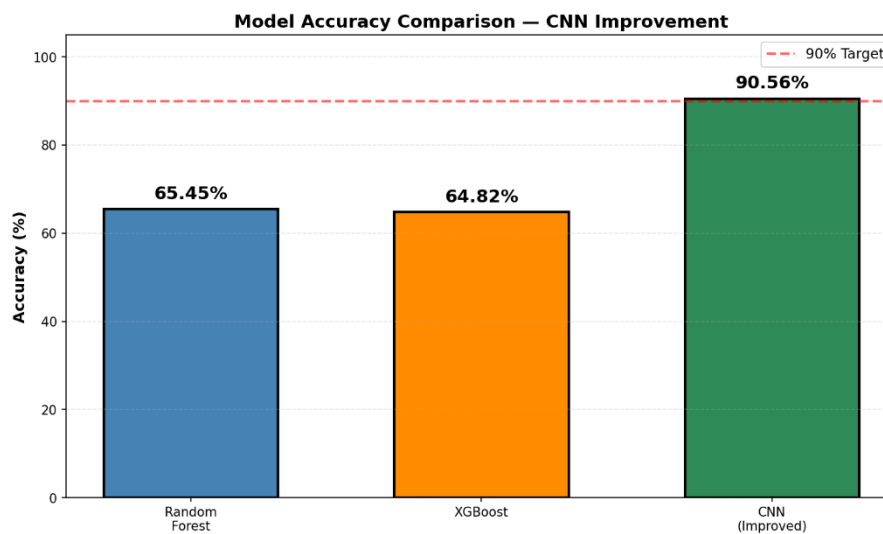


Figure 13: Model accuracy comparison — EfficientNetB0 CNN achieves 90.56%, a 25.11% improvement over Random Forest

The progression from 65.45% (Random Forest) to 90.56% (EfficientNetB0) clearly shows the benefit of deep learning for image classification tasks. The traditional models are limited by the quality of 8 hand-crafted features, while the CNN learns thousands of features directly from pixels. The EfficientNetB0 improvement over the original CNN (77% → 90.56%) demonstrates the power of transfer learning — leveraging knowledge from 1.2 million ImageNet images to classify medical MRI scans.

6.2 CNN Training Curves

The training curves show the accuracy and loss progression over all epochs (Phase 1 + Phase 2 combined, approximately 33 epochs total after early stopping).

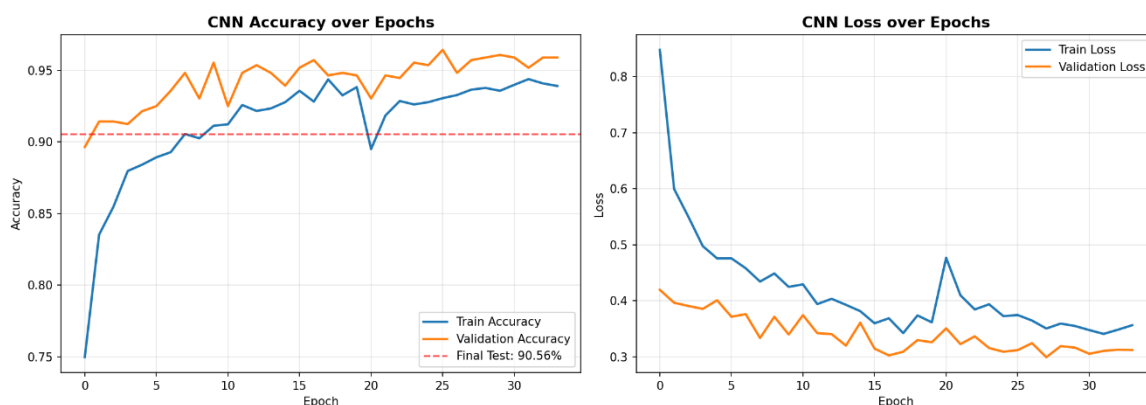


Figure 14: CNN training and validation accuracy/loss curves — both phases combined, showing stable convergence to 90.56%

Key observations from the training curves:

- The model reaches 88% validation accuracy within the first 5 epochs, showing how quickly the pre-trained EfficientNet features transfer to MRI classification.
- There is a noticeable dip in training accuracy around epoch 20, which corresponds to the transition from Phase 1 to Phase 2 (when the learning rate changes and more layers are unfrozen).
- After the Phase 2 transition, both accuracy curves stabilize and the final test accuracy (90.56%, shown as the red dashed line) falls comfortably within the validation accuracy range.
- The validation loss (orange) is consistently lower than training loss, which is a positive sign indicating the model generalizes well and is not overfitting.

6.3 Confusion Matrix Analysis

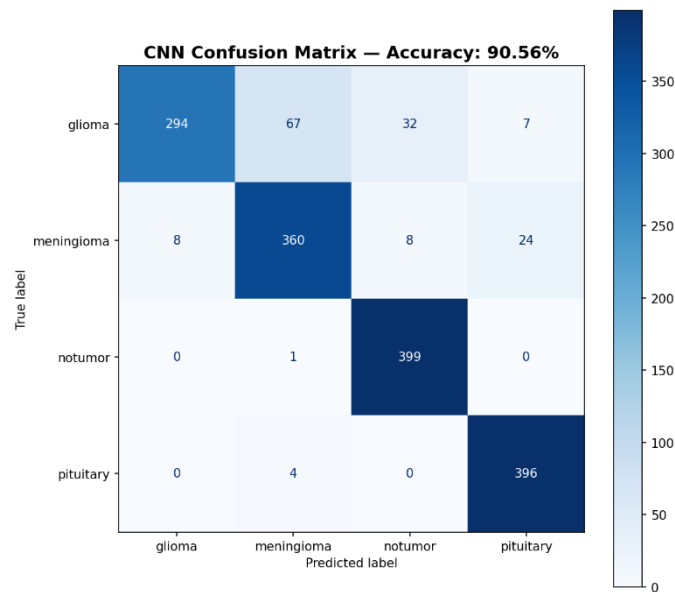


Figure 15: CNN confusion matrix — 1,600 test images, accuracy 90.56%

Class	Correct	Total	Accuracy	Main Confusion
Glioma	294	400	73.5%	Meningioma
Meningioma	360	400	90.0%	Pituitary
No Tumor	399	400	99.75%	—
Pituitary	396	400	99.0%	—

Table 2: Per-class accuracy breakdown from the confusion matrix

Detailed analysis of each class:

- **No Tumor (99.75%):** Only 1 of 400 images was misclassified. Healthy brain scans have a characteristic symmetric, regular structure with no abnormal bright mass that the model learned to distinguish very effectively.
- **Pituitary (99.0%):** Only 4 misclassifications out of 400. Pituitary tumors appear in a fixed anatomical location at the base of the brain, providing a very strong spatial cue for the model.
- **Meningioma (90.0%):** 360/400 correct. The remaining 40 errors are split between Glioma (8) and Pituitary (24), suggesting some meningiomas that appear in the vicinity of the pituitary region are confused with pituitary tumors.
- **Glioma (73.5%):** 294/400 correct, making it the most challenging class. 67 gliomas were misclassified as meningioma and 32 as notumor. Gliomas are diffuse, irregular tumors that can appear anywhere in the brain and in very different sizes and shapes — making them inherently the hardest to classify consistently.

7. Prediction Pipeline Visualization

To make the system transparent and interpretable, a full 7-step pipeline visualization was built for each test image. This shows exactly what happens to the MRI image at every stage before the final prediction is made. This kind of visualization is important in medical AI applications because it helps verify that the model is making decisions based on meaningful image features rather than artifacts.

Pipeline steps: Original MRI → Gaussian Blur → Histogram Equalization → Edge Detection → Segmentation Mask → Contour Overlay → CNN Prediction (with confidence bar chart)

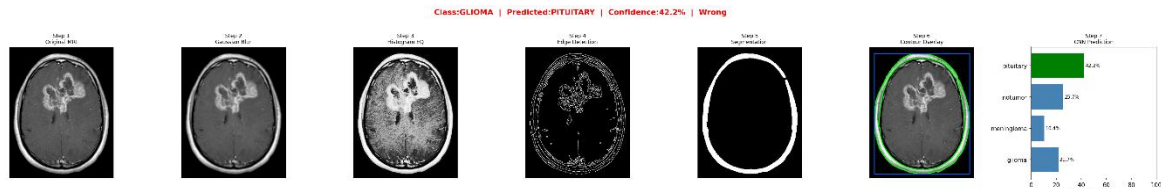


Figure 16: Glioma pipeline — Predicted: PITUITARY $\times$ (42.2% confidence). The segmentation failed to isolate the tumor, and the confidence is low (42.2%), indicating the model is uncertain — a sign the case is genuinely ambiguous.

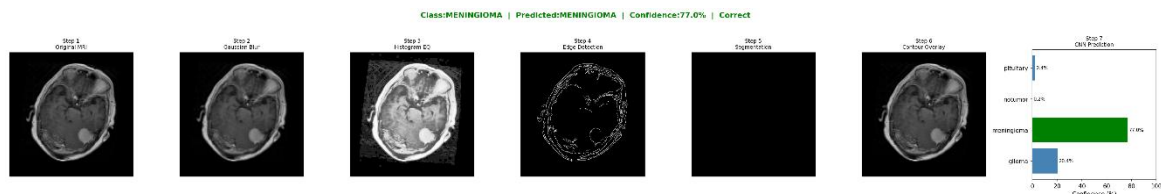


Figure 17: Meningioma pipeline — Predicted: MENINGIOMA $\checkmark$ (77.0% confidence). The model correctly identifies the tumor class despite an imperfect segmentation mask.

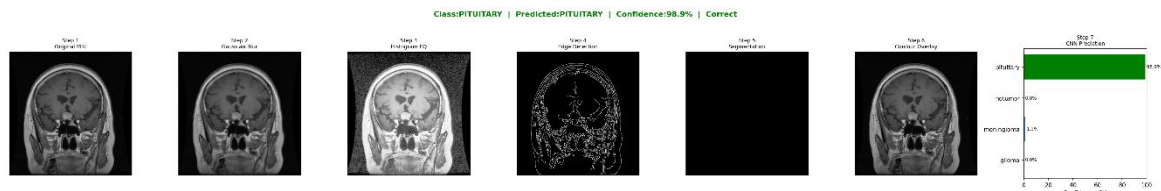


Figure 18: Pituitary pipeline — Predicted: PITUITARY $\checkmark$ (98.9% confidence). Very high confidence correct prediction — the fixed anatomical location of pituitary tumors makes this the easiest class for the model.

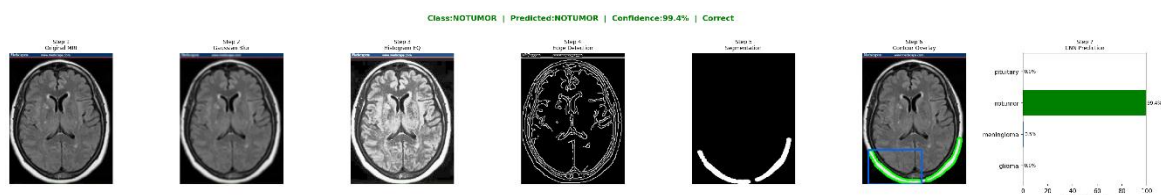


Figure 19: No Tumor pipeline — Predicted: NOTUMOR $\checkmark$ (99.4% confidence). Near-perfect confidence. The model has clearly learned to recognize the regular, symmetric structure of a healthy brain scan.

The batch prediction grid below shows 3 test samples per class (12 total) with all processing steps side by side, providing a broader view of prediction performance across different image variations within each class.

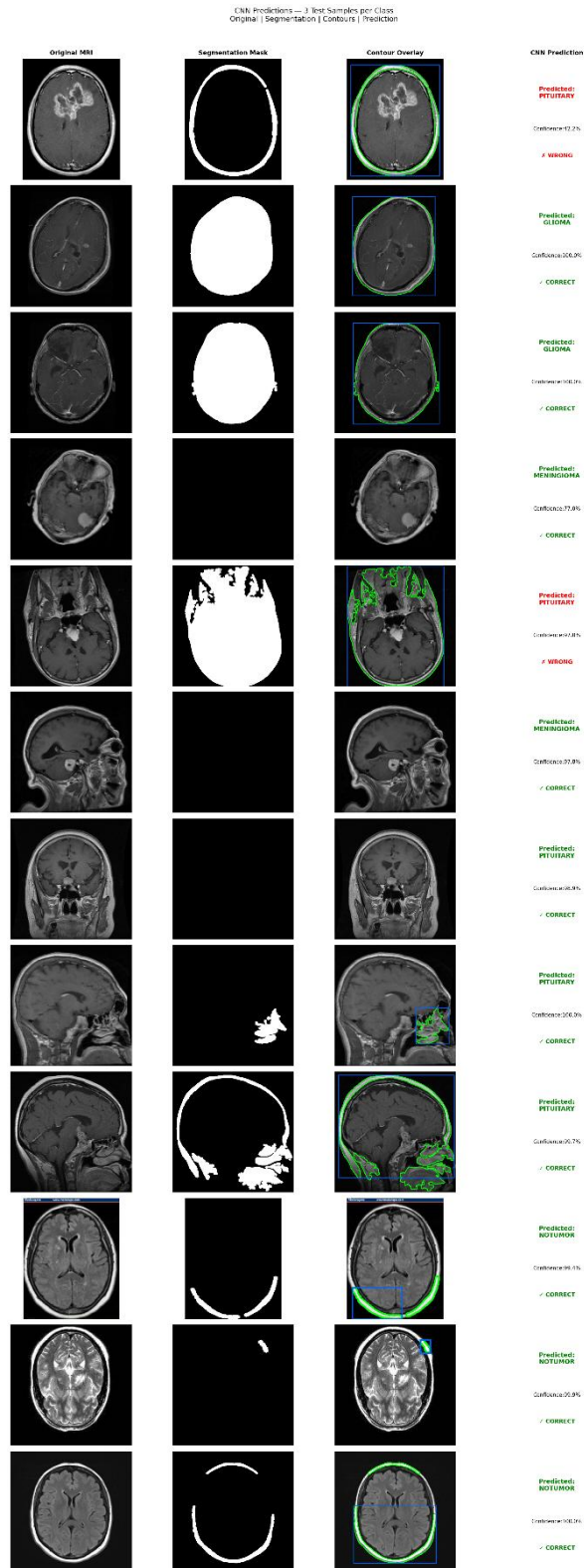


Figure 20: Batch predictions — 3 test samples per class showing Original MRI, Segmentation Mask, Contour Overlay, and CNN Prediction with confidence

8. Discussion

8.1 Key Insights

- **CNN vs Traditional ML:** The 25% accuracy gap between EfficientNetB0 (90.56%) and Random Forest (65.45%) clearly demonstrates that learning features directly from image pixels is far more effective than hand-crafted features for image classification tasks. The CNN automatically discovers thousands of discriminative features across multiple scales.
- **Transfer Learning works for medical images:** EfficientNetB0 was pre-trained on natural images (cats, dogs, cars), yet it transferred effectively to brain MRI classification. This works because the low-level features learned from natural images (edges, textures, shapes) are universal and equally relevant in medical imaging.
- **Location matters:** The two highest-scoring classes (No Tumor and Pituitary) both have strong spatial/structural characteristics that make them easy to identify. No Tumor scans have a characteristic uniform brain structure, and Pituitary tumors always appear at the base of the brain. The two hardest classes (Glioma and Meningioma) can appear anywhere and in various sizes.
- **The preprocessing pipeline helps but is not the bottleneck:** The image processing steps (blur, equalization, edge detection, morphological operations) improve image quality and produce useful features for traditional classifiers. However, the CNN's accuracy is not directly dependent on the segmentation quality — it predicts from preprocessed raw pixels. This explains why the CNN achieves 90.56% even though the segmentation is imperfect.

8.2 Challenges Faced

- **Segmentation of the tumor vs. skull:** Getting Otsu thresholding to isolate the tumor rather than the whole skull was the hardest image processing challenge. The skull boundary is often the brightest region in coronal/sagittal MRI views, so basic thresholding always detected the skull first. The fix required adding contour area filtering to skip the largest contour and only retain mid-sized blobs. Even after this fix, the segmentation still fails on images where the tumor intensity is similar to surrounding tissue — a known fundamental limitation of intensity-based segmentation.
- **Wrong preprocessing for EfficientNet:** The most critical bug encountered was using `rescale=1./255` in the data generators instead of EfficientNet's own `preprocess_input` function. This caused the model to receive completely wrong pixel values and achieve only 37% accuracy (worse than random). The fix was straightforward but the root cause was not immediately obvious — it is a subtle but critical detail when using pre-trained models.
- **Feature extraction runtime:** Running the full segmentation pipeline on all 5,600 training images was slow because each image goes through Otsu thresholding, morphological operations, and contour detection sequentially. This was addressed by first testing on 1 sample per class, then running the full loop only after confirming correctness.
- **CNN training time:** Training EfficientNetB0 at 224×224 resolution required approximately 128 seconds per epoch on CPU. The full training (both phases) took several hours. The EarlyStopping callback was essential to prevent unnecessary training time when the model had already converged.

8.3 Limitations

- **Intensity-based segmentation:** The Otsu + contour filtering approach does not work reliably for all image types. For images where tumor intensity is close to surrounding brain tissue, the segmentation mask is often empty or incorrect. A deep learning segmentation model (like U-Net trained specifically on brain MRI masks) would produce much more reliable results.
- **Limited feature set:** Only 8 features were used for Random Forest and XGBoost. A richer feature set including texture descriptors (Gray Level Co-occurrence Matrix, Gabor filter responses, Local Binary Patterns) would likely improve these classifiers.
- **No cross-validation:** The models were evaluated on a single train/test split. Cross-validation would give more reliable accuracy estimates, especially for the feature-based models.

- **Dataset diversity:** The dataset contains MRI scans from different scanners and orientations without any normalization to account for scanner-specific intensity differences. Clinical deployment would require normalization across different scanner types.

8.4 Future Improvements

- **Improve segmentation with deep learning:** Replacing Otsu thresholding with a U-Net model specifically trained on brain MRI segmentation masks would produce much more accurate tumor region isolation.
- **Train for more epochs or use larger EfficientNet variant:** EfficientNetB3 or B4 would likely push accuracy above 93% with more training.
- **Add Grad-CAM visualization:** Gradient-weighted Class Activation Mapping would visually highlight which parts of the MRI the CNN is focusing on when making its prediction — important for clinical interpretability.
- **Build a deployment interface:** A simple web application where a user can upload an MRI scan and receive a prediction with confidence scores would make the system practically usable.

9. Conclusion

This project successfully designed and implemented a complete brain tumor detection and classification pipeline using image processing and deep learning. Starting from raw MRI scans, the pipeline covered every major image processing technique: noise removal through Gaussian Blur, contrast enhancement through Histogram Equalization, structural boundary detection through Canny Edge Detection, morphological operations for mask cleanup, and tumor isolation through Otsu thresholding with contour filtering.

Three classification models were trained and evaluated. Random Forest and XGBoost, trained on 8 hand-crafted features, achieved approximately 65% accuracy — demonstrating the limitations of manual feature engineering for complex medical images. The original CNN improved this to 77% by learning directly from pixels. The final EfficientNetB0 transfer learning model achieved 90.56% accuracy, with exceptional performance on No Tumor (99.75%) and Pituitary (99.0%) classes.

The most important lesson from this project is that preprocessing and segmentation quality directly affect feature-based classifiers, but deep learning models are more resilient because they learn their own features from raw image data. Transfer learning from large pre-trained models dramatically accelerates this learning process, even when the source domain (natural images) appears different from the target domain (medical MRI scans).

The complete pipeline is transparent and reproducible — every processing step is visualized and the full prediction pipeline shows exactly how the system arrives at each classification decision. Future work will focus on improving segmentation accuracy with U-Net, expanding the dataset, and deploying the model through a clinical-grade web interface.

Project Repository (GitHub / Google Drive):

<https://github.com/Nasrullah-Hafeel/Advanced-Brain-Tumor-Detection-Segmentation-XAI>

Contents: Jupyter Notebook (.ipynb) | Output images (11 folders) | Feature CSV | Model reports | This report